ANNUAL REPORT
1988

भाकृअनुप
ICAR

NATIONAL ACADEMY OF
AGRICULTURAL RESEARCH MANAGEMENT
Rajendranagar, Hyderabad-500 030 (India)

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'The scientific and professional employees have analytical type of mind and are self-disciplined... Because of their strong achievement drives, they require recognition, status and opportunities for growth... They can be restless, sensitive and vulnerable to frustration...

High job satisfaction is the hall mark of a well managed organisation. It is fundamentally the result of effective behavioural management. Job satisfaction is dynamic. It cannot be established once and then forgotten for years. It has to be maintained week after week, month after month, year after year'.

- Keith Davis

CONTENTS

1. INTRODUCTION

1.1. Historical background :

Management of Agricultural Research is gaining increased importance for augmenting agricultural production in the country. The principles of management are increasingly applied in agricultural research, education and development as the activities related to agriculture are more and more managed on industrial lines. An institutional set up to provide training on principles of scientific management of agricultural research was felt two decades ago by ICAR and the Govt. of India. In 1969-70, the Food and Agricultural Organisation of the United Nations approached the Government of India to organize a training course in agricultural project analysis under the regional technical assistance project of the United Nations Development Programme which culminated in a Regional Training Course at Indian Agricultural Research Institute, New Delhi during November 2-28, 1970. Consequently, a National Training Centre for Agricultural Development and Project Planning was set up at IARI, New Delhi in April 1971 as a part of Fourth Five Year Plan Programme. On 28th February of 1974, the activities of National Training Centre were terminated at the end of the Fourth Five Year Plan as the scheme was self-liquidating one. The continuation of work of the National Training Centre was however felt by the authorities.

A working group under the Chairmanship of Dr.George Jacob, Chairman of the University Grants Commission was appointed during the Fifth Five Year Plan by ICAR to give concrete shape to the need for establishing a Central Staff College for Agriculture so as to ensure continuance of the activities of the National Training Centre. The group proposed the Central Staff College for Agriculture to function as a repository of ideas and information, both national and international, in the field of agricultural research and development. The proposed college was also stipulated to act as a clearing house for dissemination and utilization of such information and knowhow and to undertake a programme of publishing high quality training material and allied publications as important and priority areas of action.

In the Year 1975, details of the proposal to set up the Central Staff College for Agriculture were worked out by the sub-committee of the experts of the working group under the Chairmanship of Prof. M.V.Mathur. With a fairly wide mandate and objectives and an outlay of 50 lakhs under the Fifth Five Year Plan, the Central Staff College for Agriculture came into being on 1st September 1976. Later, from 1st April 1979, the Central Staff College for Agriculture was renamed as National Academy of Agricultural Research Management.

The Academy was started as a temporary expedient at Rajendranagar, Hyderabad where the Andhra Pradesh Agricultural Universtiy (APAU) placed certain physical facilities such as hostel, class rooms and the administrative buildings at the disposal of the ICAR. Subsequently, a broad based expert committee was appointed by the ICAR for recommending the permanent location of the Acadmey. The Committee visited several sites and locations and recommended Hyderabad as the permanent location since several infrastructure facilities and supporting institutions were available. For establishing the Academy, APAU released 50 hectares of land on lease for 99 years. On this land, the Academy constructed an Administrative Building, a Conference Hall, a Committee Room, an Academic Block, a Residence Hall with 154 rooms, a Scientist Home and quarters for all categories of staff. The Academy located at Rajendranagar is 15 kms away from the centre of Hyderabad city. The campus is spread over 50 ha. of undulating, picturesque landscape. The area is surrounded by APAU farms on Eastern, Northern and Southern sides and by National Institute of Rural Development on the Western side and is situated within Rajendranagar Municipal limits.

1.2. Objectives

The National Academy of Agricultural Research Management (NAARM) has been established with the main objectives of familiarising all those concerned with the Agricultural Research, Education and Development with the principles of scientific management in agricultural research and education. Since its inception, it has taken up the responsibility to plan, organise and conduct training programmes, workshops and seminars in the various areas related to agricultural research management for the research scientists of the ICAR and Agricultural Universities. It has also been functioning as repository of ideas and information both at national and international level and to act as a clearing house for dissemination and utilisation of such information and know-how. The Academy has also recently taken up the training programmes in the area of educational technology, curriculum development and structured learning. It has also taken up research programmes and case studies in various fields of agricultural research management and educational technology.

The main objectives of NAARM are :

1. To organise and conduct training programmes in Agricultural Research Management to Scientists at various levels.

2. To build up high quality resource material in the subject of Agricultural Research Management based on actual field experience.

3. To undertake systematic review and study of management problems of ICAR Research Institutes, Agricultural Universities, Agricultural Research Projects and Systems.

4. To organise, liaise and coordinate programmes of international cooperation in the field of Agricultural Research Management.

5. To plan, organise and conduct training programmes, workshops and seminars in educational technology, curriculum development and structured learning for the faculty of Agricultural Universities and ICAR Institutes.

The National Academy of Agricultural Research Management functions with a wide mandate. It is a novel, vibrant and emerging institution with a visionary outlook which has a dynamic approach and pragmatic policies with realistic expectations. It functions with a zest, zealousness and commitment of purpose to fulfil the mandate of the Indian Council of Agricultural Research as the major national coordinating body for agricultural research and education, and thus fulfil the national aspiration for better management of science and agriculture.

2. TRAINING PROGRAMMES

The Academy has been planning, organising and conducting various types of training programmes to meet the needs of the clientele groups working at different levels of management in the agricultural research and development organisations. Mainly three levels, viz. Junior, Middle and Senior clientele have been identified for training purposes.

2.1. Courses for Junior Clientele

Two types of courses that have been designed and offered are Foundation Course on Agricultural Research Project Management (FCARPM) and Orientation Course in Agricultural Research Project Management (ARMOC).

Foundation Course on Agricultural Research Project Management (FCARPM)

The Foundation Course for the scientists recruited to the Agricultural Research Service by the ICAR is one of the regular courses offered by the Academy. This Course aims at providing them exposure to concepts and principles of project management with special emphasis on project formulation and implementation. The Junior level scientists also receive instructions in related areas such as group dynamics, interpersonal relationships, interdisciplinary project formulation, motivation management and the like which would make them more effective research workers. The course is organised within the frame work of the following objectives:

- To acquaint the scientists with the general Agricultural Scenario in the country with reference to Research and Development.

- To present an insight into the rural society and structure.

- To equip them with the techniques in Research Project Management.

- To provide skills in organisational behaviour for high motivation, team work and conflict resolution.

- To provide training in agricultural communication and transfer of technology.

- To orient the scientists to Indian Council of Agricultural Research and Agricultural Research Service.

- To foster a feeling of fraternity among the members of ARS serving in different parts of the country.

The training programme lasts for five months. The First Phase of training includes six weeks' training at the Academy, oriented mainly to the basics of Agricultural Research Management. The Second Phase constitutes two months of training at a Regional Station of National Agricultural Research Project or ICAR Institute where they receive practical training in project identification and formulation. This phase helps the trainees develop an appreciation for rural life and environment. In the Third Phase lasting six weeks trainees return to NAARM where they are exposed to areas which will help them to become better research workers.

Agricultural Research Service (ARS) Scientists recruited through the examination conducted by Agricultural Scientists Recruitment Board (ASRB) numbering 49 scientists (XXIX FCARPM Batch) were given intensive training in various aspects of Research Project Management viz. Research Programme Planning, Appraisal, Programme Evaluation and Review Technique, Critical Path Method, and Conflict Management. Each participant was given training in the preparation and presentation of research projects on the basis of the information collected and training received during the Field Experience Training. During this training programme, the trainees were exposed also to the subjects like Agricultural Research Management, Research Project Formulation, Project Management, Organisational Behaviour, Communication, Transfer of Technology, Administrative Rules and Procedures and Practical Classes on Audio-Visual Aids.

During the year 1988, the following course contents were covered in the training :

I. **General Agricultural Scenario in the country**

- NAARM and its role
- Agro-ecological zones and cropping systems in the country
- Economics of Indian Agriculture
- Agricultural Prices
- Land Systems and measures of reform
- Agricultural Production - potential and constraints

II. **Rural Society and Structure**

- Rural Development - philosophy and programmes
- Characteristics of Indian rural life in comparison to urban life
- Rural leadership and rural development programmes
- Socio-economic surveys
- Tools of data collection for rural socio-economic surveys
- Preparation of viable research projects based on socio-economic surveys.

III. **Research Project**

- Agricultural Research Management Process
- Creativity and Problem Solving
- Research Programme Planning
- The Research Project
- Project Appraisal

- Project Selection Exercise
- Project Management Techniques - CPM & PERT
- Resource Smoothing
- Critical Path Index
- Uncertainities in Research
- Management of Research Stations

IV. Development, Transfer and Adoption of Technology

- Technology development process - promoters and retarders
- Scientists and Transfer of Technology
- Institutional Structures for Transfer of Technology
- Treatment of Research Findings
- Semantics in Scientific Communication
- Communication through Audio-Visual Aids
- Constraint Analysis
- Agricultural Communication through A.I.R.
- Agricultural Communication through Doordarshan.
- Adoption of technology - stimulants and constraints for adoption
- Development communication in Agriculture
- Transfer of Technology Experience of Developed and Developing countries.

V. History, Structure and Organisation of Agricultural Research

- History and growth of ICAR
- Structure and Organisation of ICAR
- Agricultural Research Service
- National Agricultural Research Project
- Linkages and Coordination between ICAR and Agricultural Universities.
- Linkages of ICAR with other National Research Organisations.
- Research Systems in Agricultural Universities
- Farming Systems Research
- All India Coordinated Research Projects of ICAR.

VI. Team Work and Interpersonal Relationships

- Individual and the organisation
- Groups in Organisations
- Team Work
- Group Dynamics
- Motivation and Behaviour
- Motivation and Work
- Transactional Analysis
- Conflicts in Organisation
- Positive thinking
- Case Analysis for Problem solving and decision making.

VII. Training in peripheral areas

- Communication Skills
- Rapid reading and comprehension
- Writing of Scientific Paper
- Basics of computer programmes and uses of computer in Agricultural Research
- Programming in BASIC and FORTRAN
- Energy Management in Agriculture
- Solar and Wind Energy
- Biomass and Energy
- Pollution
- Basic concepts of Linear Programming

VIII. Organisation of Research

- Agricultural Research Systems in the World
- International Research Organisations
- Research Planning and coordination in ICAR
- Contribution of ICAR to Agricultural Education System in the country
- ICAR and International collaboration in research

IX Information and Library

- Source of Agricultural Information
- Citation Indexing
- Management of personal documents
- Book reviewing

- Review of literature
- Proof reading
- Poster sessions
- Technical Report Writing.

X Trends in Research Work on some important crops/areas

- Crop Sciences
- Oilseeds Research
- Horticultural Crops
- Dryland Agriculture
- Pest Management
- Soil Management
- Fisheries Management
- Livestock Management

XI. Administration and Accounts

- General Service Conditions - An Overview
- Stores Management and Purchase Procedures
- Project budgeting and monitoring.

Orientation Course in Agricultural Research Project Management (ARMOC)

An Orientation Course in Agricultural Research Project Management (ARMOC) of one month duration was conducted for the scientists working in various institutes of the ICAR with the following objectives:

1. To discuss and acquaint with the issues relating to Agricultural Research Management process.

2. To equip and acquaint with the techniques for enhancement of research capabilities and

3. To instil the philosophy of living, thinking and working together and foster a feeling of fraternity among the members of the ARS.

The topics in this Course broadly cover the Management Process, Research Project Management, Quantitative Techniques, Organisational Behaviour, Management of Information System and Transfer of Technology.

During the Year 1988, one Orientation course on Agricultural Research Project Management for 60 S-1 inductees and promotees was conducted. The participants were exposed to various techniques of Agricultural Research Project Management.

The following course contents were covered during the training :

- Agricultural Research Management
- Project Management
- Organisational Behaviour
- Communication
- Transfer of Technology

2.2. Courses for Middle and Senior Clientele

Course on Agricultural Research Management (CARM)

In order to provide problem-centred rather than discipline-centred approaches to agricultural research, the agricultural research management should be seen in the context of development framework of carrying on basic and applied investigations, leading to the processess of development of appropriate technologies and their widespread dissemination.

The entire course is divided into following three capsules :

1. Research Project Management

2. Human Resources Management

3. Organisational Structure and Processes.

This Course on Agricultural Research Management was organised for senior research managers from SAARC countries for the first time. In all, eight senior officers and scientists from Nepal, Bangladesh, Pakistan, Sri Lanka and India attended this course which was held between August 29 and September 9, 1988.

Course on Agricultural Research Project Management (CARPM)

The Academy organised the Course on Agricultural Research Project Management (CARPM) to expose the research project managers in the ICAR Institutes to the concepts of research projects and application of the 'management approach' to the problems of agricultural research units. The course was designed to serve the interests of senior research managers heading a division in the ICAR Institutes in the cadre of S3/S4 and Professors and Heads of Departments in State Agricultural Universities. The duration of the course was two weeks.

During the year 1988, this Course was offered twice, first time from April 18-23 and second time from October 24 to November 4. The former was held solely for the senior research managers of State Agricultural Universities. 38 participants which included Associate Directors and Deputy Directors of Research, Chief Scientists and Joint Directors from 24 State Agricultural Universities attended this programme. The later programme was conducted for senior scientists and Heads of Departments of ICAR Research Institutes. 21 participants took part in this course.

The following contents were covered.

- Project Planning - Importance of planning long-term and short-term project in the management process and systematic project planning.

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Dr.N.K.Anant Rao, Former DDG (Edn) ICAR, and Dr. A.Appa Rao, Vice Chancellor, APAU, Hyderabad at the Inauguration of the Agricultural Research Management Training Programme. Also seen Dr.K.V.Raman, Director and Dr.K.V.Subba Rao, Professor.

- Project Selection - Methods and criteria used for project selection so that one can undertake better research projects in agriculture.

- Monitoring and Evaluation - Network methods for time scheduling, control and monitoring and resource smoothing of Research projects.

- Organisation Structure - Principles of organisation and the importance of technology and environment in organisation design.

- Group Dynamics - Characteristics and classification of groups, group decision making, and intergroup problems in organisations.

- Leadership - Role of leadership in management and leadership styles suitable to Indian conditions.

- Conflicts - Types of conflicts that occur in organisation and the methods used to resolve the conflicts.

- Organisational Communication - Meaning, concept and importance of organisational structure, communication behaviour, organisation structure's effect on communication, avoiding resentment, principles of effective listening, and barriers to listenig.

- Mass Communication - Television in rural communication, steps in developing a TV Programme, and rural broadcasting.

Course on Human Resources Management (CHRM)

Productivity performance of Indian Agriculture Sector has in the past relied mainly on labour-intensive methods of production. With advent of green, white and blue revolutions, labour input has been amply supported by scientific efforts. The agrarian output-mix then, crucially hinges on complex scientist-labour input-mix.

India has the largest manpower of Agricultural Scientists in the world numbering about 6480 in the ICAR system and 24,000 in the Agricultural Universities supported by an equally large technical staff. This group has to be properly motivated for meaningful researches on one hand and their optimum interaction with farmer population on the other. This involves the management of human resources (scientists, technical and supporting staff) besides financial resources.

The Academy organised the Course on Human Resources Management to expose the Research Managers to the concept of human behaviour in organisations (Organisational Behaviour) and attempt to address the issues for human resources management in Agricultural Research. The course was designed to serve the interests of the Research Managers viz. Senior Scientists heading divisions in the ICAR Institutes in the cadres of S-3/S-4 and Professors and Heads of Departments in the State Agricultural Universities. The course was offered for the duration of two weeks.

During 1988, this course was offered twice once for the senior scientists and Heads of Divisions in ICAR Institutes during August 1-12 and another for the Professors and Heads of Departments of State Agricultural Universities, during Oct. 10-21. There were 35 participants in the first and 11 in the second. In both the courses the topics covered included individual and the organisation, motivation and scientists, interpersonal relationships, transactional analysis, group dynamics, leadership, team work, conflicts, personnel policies, structure and design of organisation, organisational communication, mass communication and Research Project Management.

2.3. Workshops and Seminars

The different units of the Academy are planning, organising and conducting various types of workshops and seminars related to specialised topics in their own area of specialisation. The workshops and seminars are conducted on the basis of the research studies undertaken by them. The case materials are also developed in advance for the benefit of the participants of workshops and seminars.

During the Year 1988, four Workshops were conducted. A workshop on Curriculum Design and Development was organised for the Directors of the Summer Institutes during May 2-3 and 13 Direotors from various SAUs and ICAR Institutes participated. The workshop on Systems Agriculture in Education : Learning to deal with Complexities was attended by 30 participants from various ICAR Institutes and SAUs during July 27-30. The Workshop on Evaluation and Testing in Instructional Systems was organised during August 9-12 in which 42 persons took part in the deliberations. The workshop on User Instruction Programmes and Instructional Systems in Agricultural Universities and Research Institutes Libraries was attended by 29 Librarians during Sep. 20-24.

2.4. Consultancy Services

The Academy has been providing consultancy service in the area of Agricultural Research Management and Educational Technology. The Academy has been helping Agricultural Universities and ICAR Institutes from time to time in conducting training programmes in these areas. The Academy's Faculty are specialised in various sub-disciplines and deliver special lectures in the Institutes who arrange training programmes to their own staff. Institutes have found these lectures useful and periodically call on the services of the Faculty for this purpose.

3. STRUCTURAL ORGANISATION AND SALIENT ACTIVITIES

For the purpose of imparting effective training and undertake research studies in a meaningful way, the Academy is organised into six subject matter oriented functional units. These include

Agricultural Research Systems and Policies

Educational Systems and Technology,

Transfer of Technology Systems and Policies

Project Management,

Human Resource Development and

Information and Documentation.

All the above mentioned units function with a strong support from Centralised Services comprising

Farm Services and Agro-Technology Blocks,

Production and Publications and

Administration and Accounts.

These Units have been conducting training programmes for Agricultural Service Personnel, Agricultural Scientists from Research Institutes and Agricultural Universities, Administrators and Technology Transfer Personnel. Put together, these Units represent NAARM's Organization, (Fig.1)

3.1. AGRICULTURAL RESEARCH SYSTEMS AND POLICIES UNIT

Background

Development of Agricultural Sector is conditioned by resilience and capacities of agricultural research system. For, the functions relating to technology generation, gathering of information and procedures involved in agrarian research system would be used jointly for agricultural education, training, extension and client support to bring about desired changes in development strategies. The Agricultural Research Systems and Policies Unit studies the various systems of agricultural research in order to strengthen the various research systems that are essential for agricultural development.

Objectives, Tasks and Functions

1. To conduct systematic research in the areas of Agricultural Research Systems and Policies,

2. To organise specialised training programme in systems approach with regard to specific problems of Agricultural Research Management,

3. To evolve a package of empirically proven, reliable and valid research policies to be recommended for implementation by the authorities concerned and

4. To offer consultancy and advisory services to various research agencies for effective functioning of various reseach systems in areas like manpower planning and utilisation etc.

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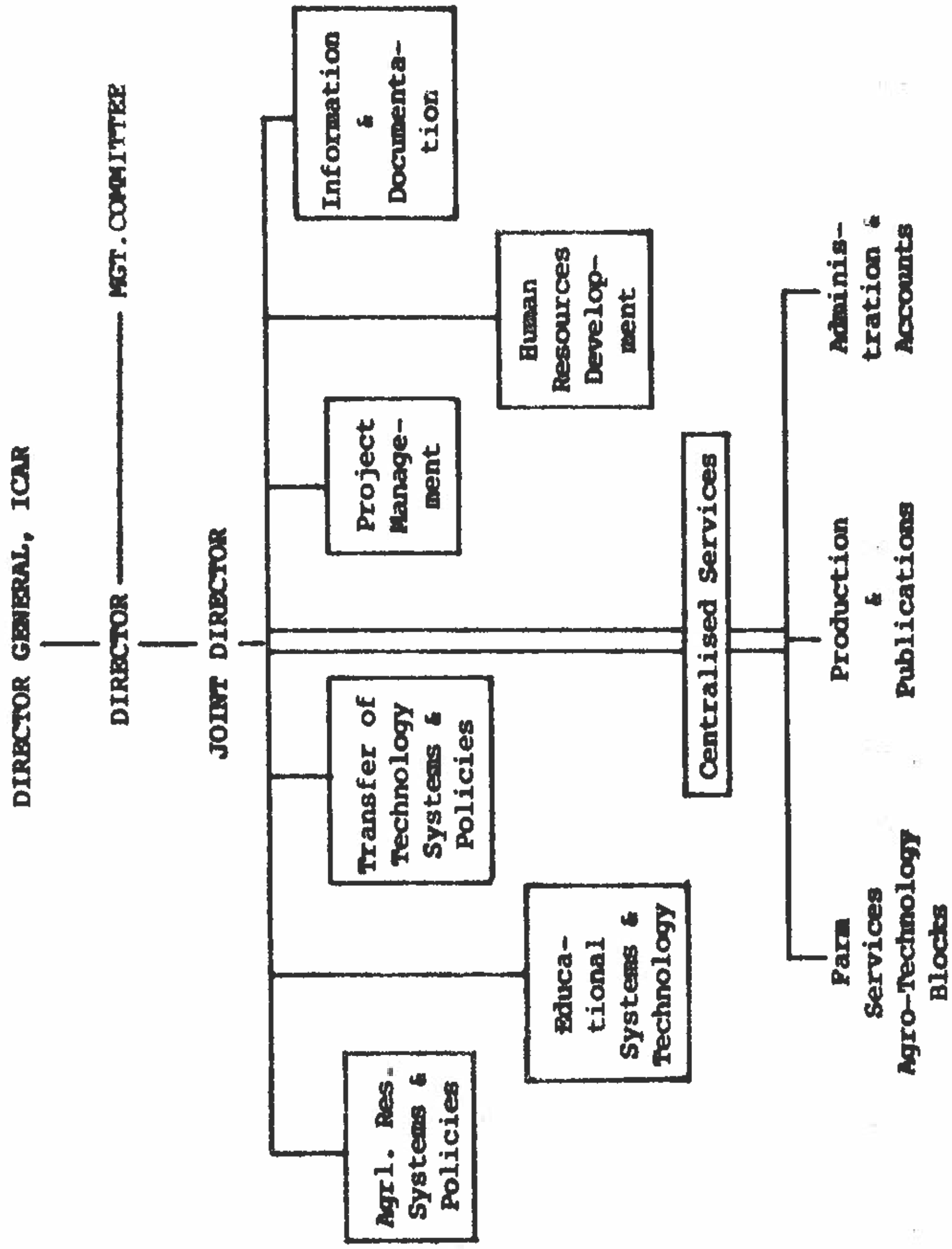

Fig.1

Activities during the Year 1988

1. A Comparative Study on the Agricultural Research Systems in the World

(T. Balaguru).

This study has been undertaken to have a global view of the Agricultural Research Systems followed in different countries, both developed and developing, including India. The study mainly focusses on the productivity of different systems as influenced by their structure and organisation so that innovations could be introduced into the system of agricultural research in India.

This study has shown that different types of research systems are being followed in different parts of the world, suiting to their specific needs and circumstances. Many research systems have been evolved over years of experience, and are constantly being restructured to suit the emerging needs. In order to meet the challenges in future, the Indian agricultural research system has to be restructured keeping in view our requirements and resources. The research sub-system at the Federal level needs to be decentralised with the emergence of regionalised research activities. The private sector would be taking up a major share of the research responsibilities in the country in future. The entire research system has to focus its attention more towards field-oriented research activities as compared to on-station activities. Above all, the existing linkages among the different components of the research system need further strengthening.

2. Farming Systems Research : A Review

(T. Balaguru).

Many developing countries are now directing their attention to a better understanding of the conditions and problems of the resource-poor small farmers to enure that research programmes and policies keep their constraints in mind. Farming Systems Research (FSR) is perhaps the most coherent approach towards this end. The present study has been undertaken to review this new approach with a view to clarify some of the conceptual issues and highlight implementation problems.

FSR represents a new direction for agricultural research because of the inability of the traditional approach to solve the problems of small farmers who work in less favourable natural environment. It essentially refers to a problem-solving research with the primary objective of improving the well being of farmers by increasing the productivity of their farming systems, given the constraints imposed by resources and the environment. It emphasizes for having multi-disciplinary groups to develop technologies consistent with farmers' indegenous technical knowledge, resources, constraints and needs, and providing for both formal and informal interaction between farmers, extension agencies and researchers. Some of the general problems encountered in the implementation of this approach are associated with designing field-oriented research, ensuring adequate resources, sustaining the interest of the researchers to engage in extensive field activities and bringing together the researchers, development agencies and farmers into a working partnership for the overall upliftment of the farming community.

1. Study on Scientific Manpower Planning

(VE. Sabarathnam)

The Project on Scientific Manpower Planning in ICAR Research Institutes was taken up with the following objectives :

(i) To trace the growth of man power resource in different disciplines in selected research institutes of the ICAR system,

(ii) To find out the relation between the objectives of the institutes and the available manpower in selected institutes,

(iii) To assess the degree of utilisation of present manpower,

(iv) To estimate the ratio between manpower availability and achievement of objectives,

(v) To work out the time lag between manpower planning and its implementation and the reason for the lag and its impact,

(vi) To predict the manpower requirement for the coming decade for selected institutes and

(vii) To elicit suggestions for effective utilisation of present manpower.

The data collected from seven different institutes covering seventy six divisions have been statistically analysed and conclusions drawn. Some of the findings which emerged out of the study are given below :

1. There was no relationship between the number of objectives and the manpower available in four out of the seven institutes studied.

2. The study failed to establish any relationship between the number of objectives and posts proposed, posts sanctioned and posts filled in the institutes under investigation.

3. A positive relationship could be established between the scientists and the number of works in which they are involved.

4. The study revealed that the number of projects did not have any bearing on the number of scientists both within and between the institutes.

5. Achievement of objectives of different projects was not found to be dependent on the availability of manpower in various divisions of the institutes studied.

6. Though proposed projects and manpower had direct relationships both within and between the institutes, such an association could not be detected at the functional division level.

7. Neither the duration nor completion of the research project was dependent on availability of manpower.

4. Interface between officials and non-officials in rural development

(VE. Sabarathnam).

A study was conducted in Madurai and Pasumpon Muthuramalingam districts of Tamil Nadu to find out the degree of interface of the officials engaged in rural development departments with the non-officials. The study revealed that majority of the respondents were having medium level interface with the non-officials. The variables which were found to have positive relationship were jurisdiction of the officials, non-officials' call on the officials at the officials' Office and memorable experiences with the non-officials. The officials suggested that wide publicity about new schemes through mass media, training the non-officials with regard to the details of a programme before its implementation, village level planning and development of non-officials for identification of beneficiaries alone would result in better interface.

5. **Measurement of perception of young scientists**
(VE. Sabarathnam)

The study on training as an experimental variable for increasing the perception of young scientists with regard to relevant research and dissemination has been continued. Preliminary analyses indicate significant differences between the perception scores obtained before and after the training. The causes of the negative perception observed in the study are also being investigated.

6. **Constraint analysis**
(K.P.C. Rao)

In order to identify the existing constraints and bottlenecks in adoption of new technologies, case studies of the following Operational Research Projects have been initiated.

i) ORP on rice based cropping systems, Khammam.

ii) ORP on rice pest control, Tenali.

iii) ORP on agriculturl drainage, Machilipatnam.

iv) ORP on Watershed development, Maheshwaram.

v) ORP on the development of SCs/OBCs, Palem.

The preliminary findings are as follows :

1. A considerable proportion of the technologies recommended by the Operational Research Pro-

jects are not being adopted by the farmers. As expected, the adoption rates were better in Operational Research Projects dealing with rice.

2. There is a need for in-depth constraint analysis which does not stop with the identification of the constraints but will undertake to test and validate the constraints perceived in each case.

3. The Scientists in the Transfer of Technology projects need to be trained with the methodologies and approaches to constraint analysis.

Course Material Developed

1. Identification of research problems through survey based rank quotient (VE. Sabarathnam).

2. Discontinuance and rejection of technology by users (VE. Sabarathnam).

3. Location specific technology development in agriculture (VE. Sabarathnam)

4. Computer concepts (K. Vidyasagar Rao).

3.2 EDUCATIONAL SYSTEMS AND TECHNOLOGY UNIT

Background.

In recent years India has made rapid strides in imparting higher agricultural education through a network of 26 Agricultural Universities, about 125 Colleges of Agriculture and allied fields and 7 ICAR Institutes. Every year a large number of agricultural graduates and postgraduates are coming out of these Institutions. Though the Agricultural Universities have inherited and adopted the educational pattern of Land Grant University of U.S.A. by incorporating integration of teaching, research and extension education functions, there are sizable inter-university variances in the instructional methods adopted in various Universities in their content and methods of imparting education and training. There is an urgent need to make higher agricultural education more meaningful and purposeful.

It was realised that though the teachers are academically qualified, many of them follow the didactic methods of teaching which do not meet the demands of practical utilisation of knowledge gathered.

Currently the country is facing new challenges in the agricultural front owing to the sudden explosion in scientific knowledge and population. Hence, there is a need to study the issues relating to the content, the structure and the function of the curriculum at the undergraduate and postgraduate levels of education offered in our Agricultural Universities and to explore the ways of making it more purposeful and responsive to the needs of changing time.

The primary function of this unit is to render assistance to Agricultural Universities and ICAR Institutes dealing with education and support for development of better teaching techniques. Modern education technology interacts closely with subject matter specialists to develop viable and useful courses and teaching techniques.

Objectives, Tasks and Functions

1. To conduct training in Education Technology as applicable to curriculum.

2. Educational planning, administration and management.

3. Instruction techniques and material production.

4. Studies on teaching-learning process including evaluation.

5. Consultancy and Advisory services.

Activities during the year 1988

Three Workshops were planned, organised and conducted during the year.

A Workshop on Curriculum Design and Development was organised from May 2-3, 1988 for 13 Directors/ ... Institutes from ICAR Institutes and Ag...

cultural Universities to help them in designing and redesigning the courses of Summer Institutes.

The Workshop on 'Systems Agriculture in Education : Learning to deal with Complexities' was organised from July 27-30, 1988 with a view to develop/modify curriculum at macro and micro levels, develop agricultural graduates so that they can manage the complexity and change, knowing the process of the development of curricula and identifying the role of experimental (problem based) strategies in higher education. Thirty Deans/Associate Deans/Directors of Instructions/Senior Professors attended the Workshop, for which Dr.Richard J.Bawden, Dean, Faculty of Agriculture and Director, Applied Systems Research Institute, Hawkesbury Agricultural College, Richmond, Australia was the Facilitator.

A Workshop on 'Evaluation and Testing in Instructional Systems' was organised from August 9-12, 1988. The Workshop was designed keeping in view the following objectives : 1. to understand the process of instructional improvement and evaluation system; 2.. to discuss techniques for measuring and evaluating instructional systems effectiveness; 3. to identify the ways and means related to improving class room instruction; and 4. to develop a testing and evaluation system for assessment of the teachers and learners.

Dr. Lawrence M.Aleamoni, Professor of Educational Psychology, Division of Educational Foundations and Administrator, College of Education, University of Arizona, Tucson, USA, was the Facilitator.

A special Computer Awareness Course for one month for the Faculty of the Academy was also conducted.

3.3. TRANSFER OF TECHNOLOGY SYSTEMS AND POLICIES UNIT

Background

Development of an appropriate transfer of technology system for effective and efficient delivery of modern information to different clientele group, therefore is **sine qua non** of productivity performance of both researcher and farmers. The effectiveness of manning the TOT systems are seen from the dimensions such as managerial efficiency of managers, span of control, communication strategies evolved and put into use, planning and execution. Further the informal and institutional linkages are the determinants of success of the TOT Systems. Finding out these linkages between and among the institutions contributing to the effectiveness of TOT systems is one of the major mandates of this Unit.

Objectives, Tasks and Functions

1. Training in Technology Transfer Processes and Appropriate Technology selection.

2. Develop effective Agricultural Communication Techniques and mechanism.

3. Identify bottlenecks and managerial deficiencies of TOT Systems and evolving more efficient processes.

4. Consultancy and Advisory Services.

Activities during the year 1988

1. **Management of Frontline TOT Projects of ICAR**

   (R.K. Samanta, R. Chittiraichelvan and A.U. Gadewar)

   The study on Lab to Land programme was continued during the year. It is proposed to take up the study of KVKs and TTCs as a continuation of this project with special emphasis on research-extension linkages.

2. **Impact of agricultural research project management training.**

   (R.K. Samanta, R. Chittiraichelvan and A.U. Gadewar)

   The Unit has taken up a study on the agricultural research project management training programmes conducted by the Academy with the following objectives :

1. To evaluate the impact of NAARM training on ICAR Scientists and their environment,

2. To find out how much has the NAARM training helped the scientists perform their jobs better in their working situations,

3. To find out how much have the client institutes benefited from their scientists trained at NAARM,

4. To document the strengths and weaknesses of existing training programmes and

5. To evaluate the training needs of the ICAR scientists.

Copies of Questionnaire (800) developed for the purpose of study were mailed to all the Institutes of ICAR and the responses of the scientists obtained. Until the end of the year, about 300 questionnaires were received. The sample population covered the NAARM trained scientists, non trained peer scientists and research managers/supervisors. After processing and analysing the data thus received, the following tentative conclusions were drawn:

1. The Agricultural Research Management training conducted by NAARM has created mixed impact among the trained and non-trained scientists of ICAR.

2. The trained scientists have better appreciation and understanding of the Agricultural Research Management training than non-trained scientists.

3. A number of new techniques and skills in different areas under Agricultural Research Management training have been learned by trained scientists.

4. Agricultural Research Management training has been able to induce team spirit, leadership and fraternity among the scientists.

5. A scientific work culture and temper has been created among the scientists through Agricultural Research Project Management training.

6. The Heads of Divisions and Directors have less appreciation about the ARPM training and the trained junior scientists.

7. The Courses Curriculum and the contents of various topics require modifications to keep up the pace and relevance with the changing needs of the scientists and institutes.

8. All categories of scientists, irrespective of their status in ICAR must undergo ARPM training at the Academy.

9. A conducive climate and infrastructure may be created in all ICAR Institutes so that the things learned in ARPM training at NAARM can be utilized and applied by the scientists.

10. The rural or field experience training for participants of FCARPM should receive higher importance and emphasis in its design, contents and implementation.

11. Due care may be taken in recruitment and development of NAARM Faculty considering the needs and suitability of the positions.

12. The existing duration of various training courses may be retained.

13. The existing evaluation system for the Foundation Courses may be modified suitably to enable continuous and effective monitoring of the trainees over the entire period of training.

14. Emphasis given in dealing with the topics like organizational development, human resource development, communication, project formulation and management and Library uses has been well received.

15. In dealing with management topics like Research Project formulation and implementation, Network Analysis and Management Process, the examples may be drawn from Indian experience with bias in agriculture.

16. The follow-up action to get feedback from the trained scientists which has not been so far attempted, may be initiated so that continuous and automatic programme modification is possible.

3. **Involvement of ICAR scientists in dissemination of research results and their perception of dissemination methods.**

(R. Chittiraichelvan and VE. Sabarathnam)

In order to find out the relationship between the involvement of the ICAR scientists in dissemination of research results and selected independent variables, a study was conducted in a few ICAR Institutes. Data collected from 300 respondents (different levels of ICAR Scientists) indicated that dissemination involvement was positively related to various independent variables viz., age, mode of ARS entry, method of evaluation of research projects (all significant at 1% level), qualification at the time of entry into service, duration of the inservice training, nature of inservice training (Indian or foreign), experience in ICAR, source of idea for the project and rules orientation nature of the institutes (all significant at 5% level).

Another dependent variable studied was the perception of scientists regarding the importance of dissemination methods in terms of ranking. The study indicated that scientists perception had positive relationships with method of project evaluation, nature of the research project, handling of Regional Research Stations by ICAR or SAUs (all at 1% level), qualifications, service satisfaction and rules orientation (all significant at 5% level).

Course material developed :

1. Development communication in Agriculture (R.K. Samanta).

2. Presentation techniques for Instructors of Library User Education Programmes (R.K. Samanta).

Instructional aids in Library User Education Programmes (R.K. Samanta).

1.4. PROJECT MANAGEMENT UNIT

Background

Project Management Unit is designed to integrate complex efforts dealing with planning, scheduling, directing and controlling of institute resources for a project that has been established with specific goals and objectives. The project management utilizes the 'system approach' to management through the use of personnel belonging to different functional disciplines assigned to a specific project. The main objective of project management is to make the most efficient and effective use of the resources of manpower, equipment, facilities, materials, money, information/technology so that the Institute objectives and goals can be achieved as per schedules, budget and the desired performance, taking into consideration the ever changing environmental input factor.

The integrated planning and control systems assure that the functional disciplines will understand their total responsibility towards achieving project needs; assure that problems resulting from scheduling and allocation of critical resources are known before hand. The problems that can jeopardise successful project completion can be forecast, so that effective corrective action can be taken to prevent or resolve the problems.

The Project Management Unit provides an individual research worker the necessary tools and techniques by which he can find proven, workable solutions to problems that can occur in the management of a project.

Objectives, Tasks and Functions

The objectives of Project Management Unit are:

1. Training in Agricultural Research Project Management Techniques

2. Case studies on planning, monitoring, controlling and evaluation of Agricultural Research Projects.

3. Network Analysis - Flow Chart and scheduling of the project activities and the networks of the projects

4. Project budgeting and financial control

5. Consultancy and Advisory services.

Activities during the Year 1988

1. **Cases for illustration on network analysis** (K. Venkateswarlu)

Several cases for illustration of ASBC, PERT, CPM, CPI, PD. and Job Boundary limits were developed. These cases have been incorporated in the book on 'Project Management Techniques' in R & D for Agriculture, Veterinary and Fishery Sciences which is nearing completion.

2. **Projectwise budgeting in ICAR Institutes.** (P. Manikandan)

The first phase of the study undertaken with the help of M/s. Ferguson & Co. to understand the existing systems has been completed. In light of the conclusions emerging out of this study, a new system is being proposed with the following features:

1. Classification of costs and establishment of budget levels based on controllability and traceability.

2. Provision for three budget levels viz. Project, Division/Service Centre and Institute.

3. Finalisation of technical programme for the ensuing year in September itself to facilitate realistic projection of budget estimates.

4. Classification of scientific equipments as project exclusive, division specific and institute level with provision for amortization

of the purchase costs over the life span of the equipments.

5. Preparation of control reports showing expenditure against budget for various levels.

6. Delegation of powers for spending to the Heads of Divisions.

The system is proposed to be introduced on an experimental basis in some of the ICAR Institutes to assess its usefulness.

Course material developed

1. Poster Presentation (P.Manikandan)
2. Project Budgeting (P.Manikandan)
3. Impact of green revolution and strategies for future (P.Manikandan)

3.5. Human Resources Development Unit

Background

People form the most important resource in any organisation. Human Resources Development Unit is concerned with methods and approaches for the growth and development of the people in organisations towards higher level of competency, creativity and fulfilment. The concept of development covers not only individuals but also groups. The Human Resources approach helps people grow in self control, responsibility and other abilities and then creates a climate in which all employees may contribute their best for improved results and efficiency. This is applicable to research organisations and scientists. Human resources development is attempted through training, feedback and counselling.

Objectives, Tasks and Functions

1. To improve the research productivity and efficiency of Agricultural Scientists and Teachers through behavioural changes by training in human resources development and organisational behaviour.

2. To conduct research in the areas of Human Resources Management and Organisational Behaviour relevant to Agricultural Research and Education Management.

3. To prepare case studies, structured experiences and role play exercises for training in human behaviour.

4. To provide consultancy in the areas of Human Resources Management and Organisational Behaviour.

Activities during the Year 1988

1. Culture of trust in research organisations
(K.Prathap Reddy, M.M.Anwer and K.V.Subba Rao)

A study on culture of trust in research organisations was initiated by the Unit with the following objectives :

1. Determining the extent of interpersonal trust/ distrust existing in different ICAR Institutes.

2. Identifying the causes for prevalence of distrust if any and

3. Arriving at some alternative courses of action for developing the culture of trust in ICAR Institutes.

The data from seven Institutes out of 18 ICAR Institutes proposed for the study have been collected and preliminary analyses of these data reveal the following :

1. The scientists by and large agreed with the existence of open communication, openness and opportunity for self control in the organisation. They also agreed that their scientist colleagues are dependable.

2. Disagreement was shown in respect of accepting others, consideration, involving others in decision making and valuing others.

3. The culture of trust was strongest at S1 scientists level and it decreased with increasing levels of hierarchy.

4. It was observed that the scientists are by and large happy with the fair administrative support, open and impartial management, fair play of their colleagues and absence of excessive politicking.

5. Two factors found to be responsible for the lack of trust were the lack of proper recognition, facilities and opportunities to do research and low professional integrety of the scientists.

3.6. Information and Documentation Unit

Background

Information and documentation form the most critical inputs in any form of management activity. Organised scientific information base is very important for decision making at all levels. Besides, a personal scientific information and retrieval system is necessary for every scientific worker to function as a more effective researcher.

Objectives, Tasks and Functions

1. To function as a repository of ideas and information, both national and international, in the field of Agricultural Research Management and Education and act as a clearing house for dissemination and utilisation of such information to the clientele institutions.

2. Training scientists in the personal scientific information management including the user education programmes.

3. To develop appropriate management information systems for scientific management.

4. To coordinate with the sectoral information centres in India in the field of Agricultural Sciences.

5. To bring out learner's information packages and transferring Workshop/Training kits.

6. Agricultural Research Management Abstracts.

7. Consultancy and Advisory Services.

Activities During the Year 1988

At present the Unit has about 19000 publications and receive 425 journals on subscription/exchange basis. The Unit has many unique features in its information storage and retrieval. Additions were continued to be made in the specialised corners viz, Foreign Tour Reports Corner, International Corner, Newspaper Clipping Service and Reprographic section.

The Reprographic section of the Unit caters to the needs of the entire Academy and during the year about 2,00,000 copies were reproduced from the plain paper copiers and 5,000 electronic scanning stencils were cut for use in the training programmes.

Agricultural Research Management Abstracts (ARMA)

This publication is brought out by the Academy for the benefit of those engaged in Agricultural Research Management, Administration, Education and Development. ARMA covers many interesting areas which come under the perview of different units of the Academy. The materials for two issues of 1988 have been released for publication. About 700 copies were printed and circulated on exchange basis.

Workshop

A Workshop on User Instruction Programmes and Instructional Systems in Agricultural Universities and Research Institutes Libraries was conducted from 20-24th of September 1988 in which 29 Librarians from Agricultural Universities and ICAR Institutes participated. The major objective of the programme was to provide the skills to the technical personnel working in Technical/Information cells for effective management of their units. Systematic maintenance of the Research Project files and adoption of modern methods in information management were also given emphasis. The programme also provided an exposure to the concepts of educationa technology and curriculum design and developmen in the library user education programmes in Indiai Agricultural Libraries. The Workshop also aime at developing a model curriculum for library use education programmes.

The following recommendations emerged ou of the Workshop deliberations.

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Participants of the Workshop on User Instruction Programmes and Instructional Systems in Agricultural Universities and Research Institutes Libraries being explained on the use of audio visual aids

1. In order to utilise the library resources effectively, the user instruction programmes have to be initiated covering study skills, research communication and related skills.

2. It was recommended that one credit compulsory course on Scientific Information Management and Research Communication should be introduced in all Agricultural Universities at graduate level.

3. For the under graduate course, the user instructional programmes may deal with the awareness of their library resources and use.

4. Learners' learning package and self instructional manuals for the users should be prepared and made available for use.

5. In order to provide the tutor Librarians, the basic skills regarding the teaching methods and the concept of educational technology, trainers training manuals and travelling workshop kits may be developed.

The Unit has taken up recently computerisation of activities like cateloguing and indexing in addition to ARMA and Library related house keeping operations.

1.7. Centralised Services

Unit of Farm Operations and Management and Agro-Technology Demonstration Blocks.

Introduction

The Academy is landscaping its campus as a model for Research and Educational Institutions. This is done, conscious of the fact that a beautifully landscaped campus leads to a comfortable environment and provides for improvement in micro-climate. The psychological comfort of green and sylvan surroundings promotes an academic atmosphere and leads to higher productivity. In landscaping the campus, stress is being laid in utilizing a large varied collection of trees, shrubs, climbers, herbaceous plants and ground covers with an eye on the educational value of a well labelled rich collection of flora.

The Agro-Technology Demonstration blocks of NAARM have been conceived as the show window of the National Agricultural Research System. The latest technology and crop varieties released by different ICAR Institutes and Agricultural Universities are being put on show here in one place (subject to agro-climate limitations) for the benefit of visitors and trainees of NAARM consisting of National and International Scientists, policy makers and administrators of Agricultural Research and Development.

Objectives, Tasks and Functions

The Farm Operations and Management is a centralised service of the Academy and has the objectives of landscaping the campus, developing and maintaining the estate and laying out the Agro-Technology Demonstration blocks in different areas of agriculture which include Field Crops, Fruit Crops, Vegetable Crops, and different Silvi systems. The landscaping activities cover the entire 50 ha. of the campus and the agro-technology demonstrations cover an area of 15 ha. of land.

Activities during the year 1988

Landscaping :

The avenues on the major roads which have been planted have been regularly maintained. These include tree species of about 20 genera. The landscaping in and around Administration Block has been completed and work is in progress in different places particularly Academic Block, Halls of Residence, Scientists Home and Faculty Block. A rose garden of over a thousand bushes have been well established. Work is in progress to establish another rose garden of Indian varieties.

Field Crops :

An area of 5 ha. has been earmarked for cultivation of various field crops, which include cereals millets, pulses, and oilseeds. 18 varieties of different crops namely maize, jowar, redgram, cowpea, wheat, sun flower, sesamum and groundnut were under demonstration. The currently recommended and latest varieties of these crops released by different ICAR Institutes, and Agricultural Universities were grown.

Horticultural Crops :

In the Orchard, six different fruit crops have been established with varieties recently released by ICAR Institutes/Agricultural Universities. The grape vineyard has been planted with 22 different varieties, guava with 5 varieties, ber orchard with 5 varieties and sapota plot with 4 varieties. Apart from these crops, work is in full progress to establish a mango orchard.

Vegetable Crops :

As a part of Agro-technology Demonstrations, vegetable crops were also demonstrated in an area of 0.5 ha. During this season vegetable crops like tomato, brinjal, chillies, bhendi, cucumber, bottle gourd, bitter gourd, sponge gourd, ridge gourd, ashgourd, cabbage, cauliflower, spinach, cluster beans, French beans, cowpea, and pumpkin were demonstrated using the latest varieties released by different ICAR Institutes and Agricultural Universities.

Silvi Systems :

Besides the conventional input systems for increased productivity, different systems of plantations have been evolved to increase the unit area productivity and to obtain food, fuel and fodder from the same piece of land. The Academy has established the following demonstrations for the benefit of the trainees and visitors coming from different parts of the country. The Silvi systems that have been established are as follows :

(i) Agro-Horticultural System : This demonstration comprises a plot of 150m x 50m planted with Balanagar variety of custard apple with a spacing of 5m x 2m and the area between the rows is being used for growing field crops like green-gram, black-gram, cowpea, etc.

This system envisages parallel production of fruit and field crops, which gives added income to the farmer.

(ii) Agro-forestry System : It is purely an agricultural system in which tree component is introduced in appropriate spacing so that the available space can be utilised for growing food crops based on the principles of mutual benefit in terms of input utilisation and association.

This demonstration is in a block of 200m x 50m with alternate rows of **Leucaena leucocephala** (Subabul variety K-8) and **Acacia auriculiformis** planted at a spacing of 5m x 2m. Both these tree species are quick growing and drought tolerant. The space between the rows is utilized for a legumenous crop.

(iii) Silvi-Pastoral System : This system has been designed to produce fodder for stall feeding/grazing purposes and supply of timber and fuel. An area of 180m x 50m unsuitable for cultivation has been planted with **Azadirachta indica** (Neem) plants at a spacing of 5m x 5m, and the complete area is sown with **Stylosanthes hamata** which is a leguminous quick growing and drought tolerant fodder crop.

This system demonstrates the effective utilisation of most unproductive land for economic benefit of farming community.

(iv) Horti-Pastoral System : It is a flexible system in which unfertile and unproductive land can be utilised for production of fruit crops and developed as a pasture either for grazing or stall feeding purposes. In this system **Emblica officinalis**, **Fesonia** sp. and **Grewia asiatica** were planted with the space between the rows sown with **Stylosanthes hamata** and **Cenchurus ciliaris** in alternate patches.

(v) Energy Plantation : A block of 100m x 50m has been planted with a recommended variety of **Leucaena leucocephala** (Subabul) at 2m x 2m spacing. This high density plantation was established to demonstrate the production of fodder and fuel in rural areas.

Green belt and graded height plantations :

Apart from demonstrating all the above systems and latest technology in field and horticultural crops, some areas particularly at boundaries of the campus were demarcated for growing graded plantation, which include tree species of eucalyptus, casuarina, tree jasmine, and silver oak. In green belt areas, different forest trees were closely planted which include tree species like Peltoforum, Siris, gulmohar, rain tree, karanj, Casia sp., silk cotton, mahwa, ashoka, neem, soapnut, drum stick, Erythrina, Spathodia, Glyricidia, Casuarina, Thespesia, tamarind, jamun, silver oak, Acacia etc.

NON CONVENTIONAL ENERGY SYSTEMS

(i) Domestic Bio-Gas Plant : A domestic bio-gas plant developed as a semi urban unit by Dr.K.G. Gollakota (Formerly of the A.P. Agricultural University) is under continuous demonstration at the Academy. In this unit non-ediable oil cakes namely castor, karanj, neem or mahua are used to produce the combustile methane gas which is fed to a burner for day to day use.

(ii) Wind Mill : A Wind Mill has been installed on the campus to harvest the wind energy for running a water pump. The wind mill has been installed with the guidance of NEDCAP. The harvested water is being utilised for irrigation of surrounding avenues and plantations.

(iii) Solar Water Heater : A multi-flat collector water heating system has been installed in the Academy for supplying hot water at 65°C. The hot water is being utilised for cooking and cleaning utensils in the kitchen room of Halls of Residence.

Production and Publications

The Academy has taken up several publications on its own to help the research managers in keeping themselves in touch with the latest techniques of management. The Academy itself publishes the materials developed by the Faculty for training purposes. Some of the publications are mentioned below :

1. Course materials and reading materials for each course conducted by the Academy.

2. Case materials prepared on the basis of research studies undertaken by the Faculty of NAARM.

GREEN AND GLORY

Green and Glory is a laboratory journal published for every Foundation Course based on the theses of the trainees of a particular batch. For every issue, 400 copies of the journal are printed and the copies are sent to all ICAR Institutes, Agricultural Universities and user agencies like various development departments of agriculture, animal sciences, dairy corporations and fisheries departments of the various states.

NEWSLETTERS

NAARM News : 'NAARM News' brings out the news and information about the events, programmes both present and future and the welfare activities of the staff of the Academy and is published every quarter.

NAARM Alumni News : 'NAARM Alumni News'is a newsletter which publishes the information for and about the Alumni half yearly.

Administration and Accounts

For providing effective administrative service this wing has been arranged in six sections viz.

1. Admn. I	Section	- deals with establishment matters
2. Admn. II	Section	- deals with Cash and Bills
3. Admn. III	Section	- deals with Stores and Purchase, Transportation and Budget matters
4. Admn. IV	Section	- deals with Diary and Despatch
5. Admn. V	Section	- deals with all training matters and compilation and publication of rules and regulations.
6. Audit & Accounts Section		- deals with all financial matters

These sections provided administrative and financial support for the smooth conducting of various training programmes of the Academy in addition to the day to day administration of the Academy.

3.8. STAFF POSITION

Name of the Post/Category	Upto 6th Plan end				Additions in 7th Plan	Total Sanctioned strength	Posts filled as on 31.12.88
	Permanent	Non-Plan	Plan	Total			
Scientific	-	14	7	21	22	43	19
Technical	3	8	2	13	25	38	32
Administrative	13	13	3	29	20	49	36
Auxiliary	5	6	-	11	19	30	17
Supporting	6	11	-	17	20	37	34
Total	**27**	**52**	**12**	**91**	**106**	**197**	**138**

l - ed ngth	Posts filled as on 31.12.88
3	19
3	32
9	36
0	17
7	34
7	138

3.9. FINANCIAL OUTLAY

(Rupees in lakhs)

Head	VII Plan Requirements			Revised Estimates 1988-89		
	Plan	Non-plan	Total	Plan	Non-plan	Total
Pay & allowances	125.10	117.52	242.62	13.86	20.77	34.63
Travelling Allowances	10.00	10.00	20.00	0.28	1.07	1.35
Recurring contingencies	25.80	111.75	137.55	8.89	21.66	30.55
Non recurring Contingencies						
Equipments	82.00		82.00	20.86		20.86
Works	277.10		277.10	105.92		105.92
Others		10.73	10.73	0.19		0.19
Total	**520.00**	**250.00**	**770.00**	**150.00**	**43.50**	**193.50**

4. UNDP ADVANCED CENTRE ON AGRICULTURAL EDUCATION AND RESEARCH MANAGEMENT

An Advanced Centre on Agricultural Education and Research Management has been established at the Academy with assistance from UNDP/UNESCO/FAO with the primary objective of establishing a system for managing agricultural education and research so as to ensure continuous adjustments to changing national needs through scientific study of educational research management process and training key personnel in their application.

The centre conducted one Workshop on Curriculum Design and Development, one each on Systems Agriculture in Education : Learning to deal with complexities, and Evaluation and testing in Instructional Systems, one course on Agricultural Research Management for the participants from SAARC Countries, two courses on Human Resources Management and two courses on Agricultural Research Project Management.

During the Year Dr. R.H. Maxwell, Head of the Division of International Agriculture and Forestry West Virginia University, Morgantown, USA, Key Consultant to the Centre was at the Academy between January 25 and February 4, 1988 and held discussions with the Director and the Faculty members of the Academy. He had tendered advice on planning and organisation of a number of training programmes/workshops by the Centre and also procurement of various modern equipments

Dr. Richard John Bawden, Dean, Faculty of Agriculture and Director, Applied Systems Research Institute Hawkesbury Agricultural College, Richmond, Australia another Consultant to the Centre was at the Academ between 9 July and 8 August 1988. He was the facilitato for the Workshop 'Systems Agriculture in Educatio : Learning to deal with Complexities' held from Jul 27-30, 1988.

Another Consultant, Dr. Lawrence M. Aleamoni Professor of Educational Psychology, Division of Educa tional Foundations and Administration, University c Arizona, Tucson, USA held discussions with the Facult (July 23 to August 22, 1988) of the Educational Systen and Technology Unit of the Academy and also was tl Facilitator for the Workshop 'Evaluation and testir in Instructional Systems' held from August 9-12, 198

Dr. Devesh Kishore, Professor and Head, Educational Systems and Technology Unit returned to the Academy on March 2nd 1988 after six months of advanced study in educational administration and management at University of Arizona, Tucson; Syracuse Univ., New York; Cornell University, Ithaca, New York and Ohio State University, Columbus, USA, under a fellowship awarded by the Centre.

Dr. R. K. Samanta, Associate Professor and Head, TOT Systems and Policies Unit joined the Academy on March 2, 1988 after a training of six months under a fellowship of the Centre on Audio Visual materials production at Cornell University, Ithaca, New York; Syracuse University, New York and Ohio State University, Columbus, USA.

Dr. V. Raghavendra Rao, Professor in Project Management Unit left for USA for advanced study in the field of Project planning and prioritisation for a period of six months from July 20, 1988.

Dr. T. Balaguru, Associate Professor in Agricultural Research Systems and Policies Unit also left for USA on July 30, 1988 for a period of six months to get advanced training in the field of Research Systems and Research Methodology.

Dr. K. Venkateswarlu, Professor in the Project Management Unit left for USA on December 11, 1988 for advanced training in Network analysis in the area of Agricultural Research and Project Management under this project for a period of six months.

D. PHYSICAL FACILITIES

The Academy has set up the following laboratories to make the teaching-learning process more effective and help the trainees in effective use of mass media for dissemination of information.

D.1. Closed Circuit Television Systems Laboratory:

It provided intensive training to scientists in preparation of effective and intelligible material for telecasting and to conduct role-play exercises for personality development. The laboratory was also useful for developing proper speech communication skills and to improve the group communication process in building an efficient research team. The lab provided with modern video cameras also produced educational programmes related to structured learning.

5.2. **Audio-Visual Laboratory including Photography Unit:**

This lab provided an excellent opportunity for scientists to have first-hand communication and practical experiences in preparation and use of A V aids for effective presentation. This laboratory has the facilities for audio dubbing and has several audio visual equipments including overhead projectors, Epidiascopes, coloured transparency making units, slide projectors, 16mm projectors, tape recorders and public address systems.

5.3. **Speech and Mass Communication Laboratory:**

It helps the Scientists to record their programmes such as interviews, discussion and talks specially from the rural listeners' point of view.

5.4. **Organisational Behaviour Laboratories:**

These laboratories are organised for exercise in various aspects of organisational behaviour and help the trainees to understand human thought, action and behaviour.

5.5. **Self-Instructional Carrels**

These carrels are being established at the Academy to provide an opportunity to the participants of various training programmes, workshops, seminars and symposia in self-learning process. A slide projector, a video cassette player with monitor, micro fisch reader and an overhead projector are made available in each carrel for this purpose.

The scientists are being trained in the use of these equipments for effective presentation of scientific material.

All the lecture halls of the Academy, the Committee Room and the Conference Hall are provided with the necessary audio-visual equipment to make the lecturing process more efficient and effective.

5.6. Computer Facilities

The Academy is provided with one Micro computer of UPTRON S-850 along with the Printer Centronics [illegible] of 200 cps. It has the facilities such as the Data Analysis, Dataform, Graphics and Word Processing. The section also acquired a PC-AT of DCM with 1 MB System memory and one 40 MB Hard Disk drive along with a L & T Printer having a speed of 160 cps. These Computers are mainly used for Data Analysis of Research Projects and for Training. Further, they are also used for demonstration purposes in exposing the functions and operations of the PCs and Micro computers to the ARS Trainees.

Steps were initiated during the year to procure a mainframe computer of WIPRO with 11 intelligent terminals along with two dot matrix printers, one line printer and one plotter.

In addition, the computer cell has undertaken the following works during the year:

i) Rendered assistance in the preparation of questionnaires and schedules for data collection on various projects.

ii) Development of programmes for analysis of data.

iii) Documented the Research Projects, Reports and Research Papers of various Units, Bio-data of trainees who attended the Foundation Course during the year, and addresses of the members of the NAARM Alumni.

iv) Prepared the pay bills and pay slips of the Officers.

v) Bio-data of Faculty Members of NAARM have also been documented for ready reference.

5.7. Banking Facilities

Central Bank of India, Golconda Fort Branch, Hyderabad continued to operate an Extension Counter (regular) at NAARM to cater to the banking requirements of Academy, Faculty Members, Staff and Trainees.

5.8. Capital Works

The construction of Faculty-cum-Laboratory building is in full progress. The front block is nearing completion and the work of rear block is in full swing. In addition, construction of 66 residential quarters has commenced. A tractor vehicle shed adjacent to Administrative Block is nearing completion.

6. NAARM ALUMNI ASSOCIATION

The total membership of the Association increased to 520 as on 31.12.88. Elections were held for the Office bearers of the Association in June 1988 and the new Council of Office bearers has taken over for a period of two years. The President also nominated a new General Secretary. The Association decided to broaden the areas of activities and the following programmes have been formulated for execution during the next year.

i) Organisation of NAARM Alumni Association Foundation Day Lectures by eminent persons.

ii) Participation of Alumni in a larger measure in the publication of Alumni Newsletter.

iii) To design a logo/emblem for the Association.

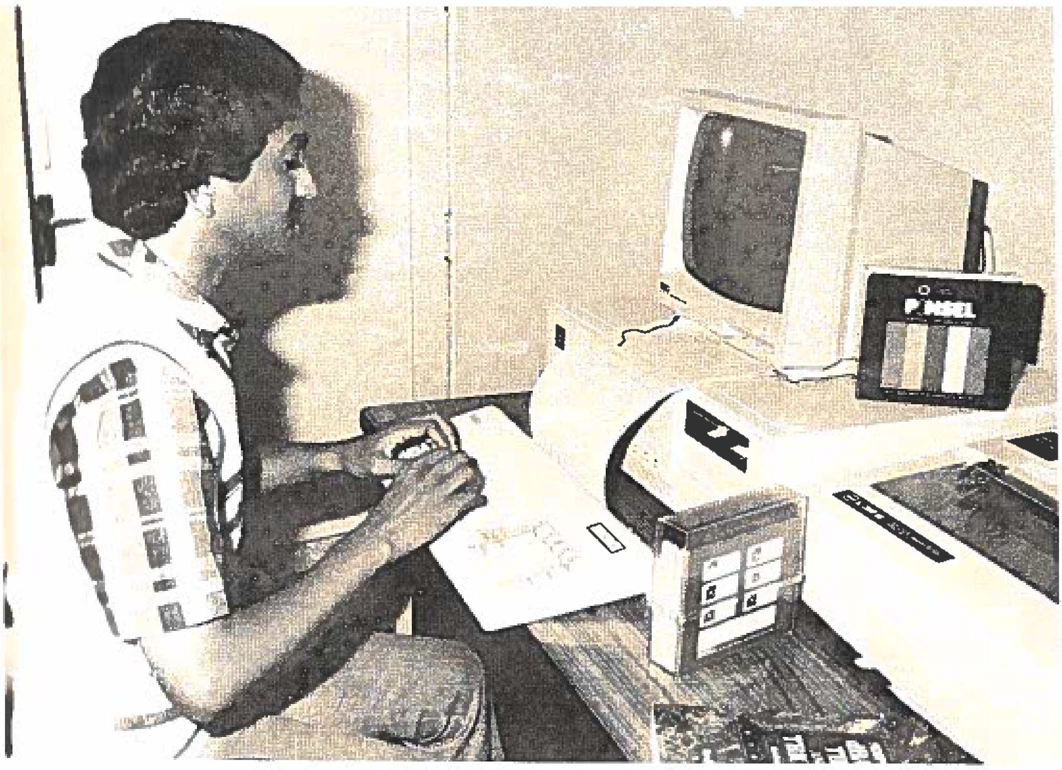
The Association has brought out two issues of NAARM Alumni News during the year. The Association has also been permitted by the Govt of India to accept foreign contributions under the relevant rules.

7. PUBLICATIONS

7.1. Scientific Papers Published

1. Balaguru T., Venkateswarlu K., and Rajagopalan M., 1988. Implementation of World Bank aided National Agricultural Research Project in India : A case study. Agric. Admn & Entension, 29 : 135-47.

2. Devesh Kishore 1988. Media's Role in modernising Agriculture. Vidura, 25(5) : 6-8.



Work on Computer in progress

3. Pestonjee D.M., and Prathap Reddy K. 1988. Development of a Psychometric measure of learned helplessness (LH). Working Paper No.746, IIM, Ahmedabad.

4. Raman K.V. and Balaguru T. 1988. NARP - An innovative approach towards FSR in India. Agric.Admn &Extension, 30 : 203-213.

5. Sabaruthnam VE. 1988, Youth forces for greening dryland. Intensive Agriculture 16(8) : 7-10.

7.2. Books Published/Edited

1. Raman K.V., Anwer M.M. and Gaddagimath R. B. (Ed). 1988. Agricultural Research Systems and Management in the 21st Century, NAARM Alumni Association, Hyderabad.

2. Raman K.V., Balaguru T., and Manikandan P. 1988., National Agricultural Research, Education and Extension Education Systems in India. National Academy of Agricultural Research Management, Hyderabad.

3. Venkataratnam L., Anwer M.M., Chittiraichelvan R., Satyanarayana G. and Rameshwar A. (Ed.) 1988. Packaging of Fruits and Vegetables in India, Agri-Horticultural Society (Regd), Hyderabad.

7.3. Papers presented at Symposia/Seminars/Workshops/Conferences

1. Balaguru T. Agricultural Education System in India. Presented at the Agricultural Education Graduate Committee Meeting of the Ohio State University, Columbus, Ohio, USA, October 21, 1988.

2. Devesh Kishore. Curriculum Planning. Presented in the National Conference of All India Association for Educational Technology at Goa. Dec. 22-24, 1988.

3. Gadewar, A.U. and Chittiraichelvan R. Women in changing Agricultural Scenario : A critical review. Presented at the Seminar on Role of Farm Women in Agriculture held at PKV, Akola, October 7-8, 1988.

4. Raman K.V., NAARM and its role in promoting excellence in Agricultural Education and Research Management. Presented at the Indian Agricultural Universities Vice Chancellors' Convention held at Trivandrum, December 16-17, 1988.

5. Raman K.V., and Balaguru T. NARP as a modified FSR approach in India. Presented at the International Symposium on FSR/E held at the University of Arkansas, Fayeteville, USA, Oct. 9-12, 1988.

6. Rao, K.P.C. Institutional Credit and Dryland Farmers - A study in Gadwal Block of Mehboobnagar Dist., (A.P.). Presented in 48th Annual Conference of the Indian Society of Agricultural Economics at Banaras, December 27-29, 1988.

7. Rao, K.P.C. Social and Financial constraints for tree based cropping systems. Presented in Alternate Land Use Systems in Dryland Agriculture at CRIDA, Hyderabad, November 14-26, 1988.

8. Rao, K.P.C. Development of suitable proforma and interview methods for collection of data on socio economic factors presented to the working group meeting on socio-economic factors for development of aternate land use systems in Arid/Semi arid regions held at CRIDA, Hyderabad on October 11-12, 1988.

9. Sabarathnam VE. Interface index of rural development officials with non-officials in Tamil Nadu. Paper presented at the Seminar on Organisational Dynamics for Rural Development held at NIRD, Hyderabad, July 11-14, 1988.

10. Samal, P. and Rao, K.P.C. Strategies to stablise oilseed prices in India. Presented at the National Seminar on Strategies for making India self reliant in vegetable oils at Hyderabad, September, 5-9, 1988.

11. Samanta, R. K. and Prathap Reddy K. Organisational development vis-a-vis rural development programmes. Paper presented at National Seminar on Organisational Dynamics for Rural Development at NIRD, Hyderabad, July 11-14, 1988.

7.4. Special lectures delivered

Dr. Devesh Kishore

* Management of Agricultural Development Pro

jects with special reference to application of science and technology, on April 7, 1988 at NIRD, Hyderabad.

* Organisational structure, climate and communication as vehicles for effective human resources development, to the participants of the 4th Training Seminar at SVP National Police Academy, Hyderabad on April 5, 1988.

* Use of Video technology as innovative method of class room teaching to the participants of Summer Institute on Educational Technology for the Teachers of Agricultural Universities at APAU, Hyderabad on June 20, 1988.

* The guidelines for modification of UG and PG Curriculum and courses with the help of case studies and models, Ibid, June 28, 1988.

* Organisational Communication at State Bank Institute of Rural Development, Hyderabad on August 8, 1988.

* Application of educational technology to teaching learning process, at St. Annes College of Education, Secunderabad on August 25, 1988.

* Forms of science and technology communication, at Dept. of Agricultural Extension and Rural Sociology, TNAU, Coimbatore on December 8, 1988.

Dr. H. K. Samanta

* Instructional aids and effective teaching learning situation, at St. Annes College of Education, Secunderabad on August 25, 1988.

* Development Communication and Technology Transfer, at Directorate of Rice Research, Hyderabad on September 20, 1988.

* Media support to T & V System in Agricultural Development, at Extension Education Institute, APAU, Hyderabad on September 20, 1988.

* Development Communication in Agriculture, at State Institute of Rural Development, Hyderabad on December 10, 1988.

Dr. VE. Sabarathnam

* Location-Specific technology development at Agricultural College and Research Institute (TNAU), Madurai on June 7, 1988.

* Integrated Communication for animal husbandry development, at State Institute of Rural Development on November 16, 1988.

Dr. K.P.C. Rao

* Govt. of India Programmes for Rural Development, at State Bank Institute of Rural Development, Hyderabad on December 14, 1988.

8. MISCELLANEOUS

8.1. Faculty Development

** Dr. K. V. Raman, Director attended a Workshop on Farming Systems Research as Chief Guest at Birsa Agricultural University, Ranchi on February 18 - 19, 1988.

** Dr. K. V. Raman, Director took part in the meeting of the Vice Chancellors of Agricultural Universities to discuss the report 'Indo-US Impact Evaluation of Indian Agricultural Universities' held at Delhi on May 21-22, 1988.

** Dr. K. V. Raman, Director took part in the deliberations of the inter-disciplinary scientifi panel of NARP on August 18-19, 1988 at ICAR New Delhi.

** Dr. K. V. Raman, Director attended the Annua Convention of Clay Minerals Society of Indi held at Orissa University of Agriculture an Technology, Bhubaneshwar on November 22-23, 198

** Dr.Devesh Kishore, Professor, participated i a preparatory workshop to decide on curriculu development and training methodology for th Course on Extension Management for Farm Wome held at MANAGE, Hyderabad, on March 12, 1988.

- Dr. R. K. Samanta, Associate Professor attended a Seminar on Organisational Dynamics for Rural Development at NIRD, Hyderabad from July 11-14, 1988.

- Dr. R. K. Samanta, Associate Professor attended a workshop on Rapid Rural Appraisal at NIRD, Hyderabad from Oct. 27-28, 1988.

- Dr. VE. Sabarathnam, Associate Professor participated in a workshop on Management of Research in Rainfed Regions from December 13-15, 1988 held at CRIDA, Hyderabad.

- R. Chittiraichelvan, Associate Professor participated in the National Seminar on Dryland Horticulture held at CRIDA, Hyderabad from July 20-22, 1988.

- Dr. M.M. Anwer, Associate Professor attended the National Seminar on Dryland Horticulture at CRIDA, Hyderabad from July 20-22, 1988.

- Shri G.C.Sharma, Sr.Administrative Officer attended a programme on Computer Education and Training conducted by Computer Maintenance Corporation Ltd., at Hyderabad from December 28, 1987 to January 8, 1988.

8.2. Foreign Visits

- Dr. K. V. Raman Director visited South East Asian Countries from March 14-31, 1988 for arranging IDRC Workshop in collaboration with NAARM. He visited MADRI, Oil Palm Research Institute, Rubber Research Institute, Agricultural Universities and Forest Research Institute at Kuala Lumpur; the FAO Representative and the Director General of Agricultural Research and his senior staff at Jakarta, University of Philippines, [illegible], SEARCA and FCARRD. He also visited the FAO Regional Office at Bangkok, Rector and [illegible] Kasetsart University, the Governor of TISTR [illegible] the Director General of Agriculture.

* Dr. K. V. Raman, Director, visited USA for participating in Rutgers/ISNAR Agricultural Technology Workshop held from July 4-9, 1988.

* Dr. K. V. Raman, Director participated in USAID sponsored International Workshop on Invitation to dialogue : The role of the Agricultural University in the 21st Century from October 2-8, 1988 held at Sheraton International Conference Centre at Reston, Virginia, USA.

* Dr. K. V. Raman, Director took part in the deliberations of ISNAR's Annual International Research Management Workshop on Human Resource Management in National Agricultural Research held from November 7-11, 1988 at the Hague, Netherlands.

* Dr. K. V. Raman, Director, attended the first meeting of the Governing Board of the SAARC Agricultural Information Centre at Dhaka from December 4-5, 1988.

* Dr. Devesh Kishore, Professor and Head, Educational Systems and Technology Unit returned to the Academy on March 2nd, 1988 after six months of advanced study in educational administration and management at University of Arizona, Tucson; Syracuse University, New York; Cornell University, Ithaca, New York and Ohio State University, Columbus, USA under a fellowship awarded by the UNDP Centre.

* Dr. R. K. Samanta, Associate Professor and Head, TOT Systems and Policies Unit joined the Academy on March 2, 1988 after a training of 6 months under the fellowship of the UNDP Centre on Audio Visual materials production at Cornell University, Ithaca, New York; Syracuse University, New York and Ohio State University, Columbus, USA.

* Dr. V. Raghavendra Rao, Professor in Project Management Unit left for USA for advanced study in the field of project planning and prioritisation for a period of six months from 20th July, 1988.

- Dr. T. Balaguru, Associate Professor in Agricultural Research Systems and Policies Unit also left for USA on July 30, 1988 for a period of six months to get advanced training in the field of Research Systems and Research Methodology.

- Dr. K. Venkateswarlu, Professor in the Project Management Unit left for USA on December 11th, 1988 for advanced training in Network Analysis in the area of Agricultural Research and Project Management for a period of six months.

Distinguished Visitors

- Hon'ble Union Minister for Agriculture, Shri Bhajan Lal visited the Academy on June 2, 1988. Dr. K. V. Raman, Director accorded a warm welcome to the Hon'ble Minister and introduced the Faculty members. Dr. Devesh Kishore, Professor, explained the activities of NAARM to the Minister. An exhibition highlighting the various activities of the Academy was arranged on the occasion.

- A twelve member delegation from Arab Countries visited the Academy on March 10, 1988. The team consisted of

  1. Dr. Amer E. Megri, Libya.
  2. Dr. Abdel Moneim B. El Ahmadi, Sudan.
  3. Mr. Hamza Nackur, Tunisia.
  4. Mr. Bamakhramah H. S. PDR Yemen.
  5. Mr. Mellas Hamdoun, Morocco.
  6. Mr. Zouttane El. Madani, Morocco.
  7. Dr. Ahmed Hassan Nour, Egypt.
  8. Dr. Adnan Swaid, Syria.
  9. Mr. Benbouzid Ahmed, Algeria.
  10. Dr. Khaldi Gley, Tunisia.
  11. Mr. Hassan Al Ahmad, Syria.

12. Dr. Mabruk Shtaiwi Turki, Libya.

Dr. K. V. Raman, Director accorded a warm welcome to the delegates and explained the activities of NAARM. He also gave an overview of the National Agricultural Research System. The delegation expressed the hope that the Academy will be able to provide support in agricultural research management and educational technology to the Arab countries.

* Meena Datta, Programme Specialist USAID, New Delhi visited the Academy on April 8, 1989.

* Ronald H. Pollock, Agricultural Research Office USAID,New Delhi visited the Academy on April 8, 1988.

* Dr. Albert L Beown, Dy. Director, Chemonics International, Washington DC visited the Academy on April 8, 1988.

* Dr. Dónald Barton, Chemonics Midterm Review (Retd.) Cornell University, New York visited the Academy on April 8, 1988.

* A three member delegation from Chinese Academy of Agricultural Sciences, Beijing, China visited the Academy on April 30, 1988. The members were:

  1. Prof. Liu Zhicheng, Vice President.
  2. Prof. Yang Yan, Project Officer, Foreign Affairs Division.
  3. Prof. Xin Naiquan, Dy. Director, Scientific Management Department.

* A Three member delegation from Burma visited the Academy on June 30, 1988. The members were:

  1. Mr. U. Myat Twe, Dy. General Manager (Fibre Crop Division), A.R.I., Yezin, Pyinmana, Burma.
  2. Mr. U. Tun Hlaing, Dy. General Manager, A.R.I. Yezin Pyinmana, Burma.
  3. Mr.U. Tin Hla, Project Manager, Agriculture Corporation, Rangoon, Burma.

Honourable Union Minister for Agriculture
Shri Bhajan Lal at the Academy

* Dr. P.S.O. Okoli, Federal Agricultural Coordinating Unit, Nigeria visited the Academy on August 6, 1988.

* Mr. A. A. Pianchi, Director-General, Kundoz, Agril. Res. Station, Republic of Afghanistan visited the Academy on August 30, 1988.

* Mr. Nooruddin Basier, Min. of Agric. Kabul, Afghanistan visited the Academy on August 30, 1988.

* Dr.M.S.Swaminathan, Chairman, IUCN visited the Academy on September 14, 1988.

* Prof. Mrs. (Betty) Laverne Forest, University of Wisconsin Madison, Wisconcin visited the Academy on September 15, 1988.

* A team of seven Agricultural Directors/Chief Scientific Research Officers from SADCC countries visited the Academy on September 16, 1988. The team consisted of:

1. Daniel Lilborio De.Sousa, Director of Agricultural Research, Min. of Agric. Maputo, Mozambique.

2. Dr. Robert Kikopa, Acting Director of Research and Training Dar-es-Salaam, Tanzania.

3. D.B.R. Manda, Chief Agril. Res. Officer, Min. of Agriculture, Lilongwe 3, Malawi.

4. Maseabata Ntoanyane, Acting Chief Agric. Research Officer, Farming Systems Project, Deptt. of Agricuture, Maseru, Lesotho.

5. Mr. Isaac Clmlko Nk hungulu, Asst. Director, Extension, Min. of Agriculture, Lusaka, Zambia.

6. Christopher T. Nk wanyane, Chief, Research Division, Agric. Res. Division, Swaziland.

7. Dennis M. Wanehinga, Manpower & Training Officer, SADCC, Botswana.

* Earl H. Teeter Jr, Prog. Leader, Incharge, International Programmes, USDA, Washington DC visited the Academy on September 19, 1988.

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* Le Quang Buu, Rice Research Institute of Mekong Rice Delta, OMON, Haugiang, visited the Academy on October 17, 1988.

* Ngyen Minhtri, Vietnam Embassy, visited the Academy on October 17, 1988.

* Dang Kim Son, National Rice Research Institute, Hangiang Province, Socialist Republic of Vietnam visited the Academy on October 17, 1988.

* Dr. P. Selvie Das, Chairman and Management Committee, Regional College of Education, Mysore (NCERT) & Vice Chancellor, University of Mysore visited the Academy on October 28, 1988.

* Osman Aballa Mohamed, Dy.Director, Agric. Res. Extension and Training Project (ARETP), Amarat, Khartoum, Sudan, visited the Academy on November 16, 1988.

* Ronald. E. Ostman, Associate Professor, Communication, Cornell University, Ithaca, New York visited the Academy on December 8, 1988.

* Dr. Padma Ramachandran, Institute of Management in Govt., Trivandrum visited the Academy on December 30, 1988.

8.4. Honours and Awards

In the 48th All India Industrial Exhibitions' Horticultural Show, the Academy won three First Prizes for the best collection of grapes and best exhibits of Cut Flowers. The Academy was also given nine second prizes for different categories of exhibits of fruits and vegetables. Two special prizes were also given for collection of Bougainvilleas and Cactii.

In the Rose Show, the Academy has won the prestigeous Challenge Trophy for the best display of roses. Five first prizes, three second prizes and four third prizes were also won for different categories of roses by the Academy.

8.5. Important Committees of the Academy

Management Committee

The Academy is guided in planning, organising and conducting of various types of training programmes

Dr.M.S.Swaminathan meeting the Faculty Members
at the Academy

and research studies by the Management Committee of NAARM. The Management Committee approves the programmes to be undertaken. The Academy ensures their implementation both in terms of physical targets and schedules and provides broad guidelines for future programmes. The Management Committee also helps in formulation and consideration of proposals for Five Year Plan and Annual Plan and takes up periodical review of progress of development schemes.

The Management Committee besides assisting Director of the Academy in the discharge of his functions particularly concentrates his attention on the Research and other programmes of the Academy and ensures their implementation. While examining these programmes periodically, it pinpoints bottlenecks, if any, and suggests suitable remedial measures.

The 11th and the 12th meetings of the Management Committee of the Academy were held on January 1, 1988 and June 25, 1988 respectively. The composition of the Management Committee of NAARM is as follows :

1.	Dr. K. V. Raman, Director, NAARM, Rajendranagar, HYDERABAD - 500 030	Chairman
2.	Dr. Hanumanth Rao, Director, Animal Husbandry, Govt.of A.P. Santinagar, HYDERABAD - 500 028.	Member
3.	Director of Agriculture Govt. of Punjab, LUDHIANA - 141 001. (Punjab)	Member
4.	Dr. A. Appa Rao, Vice-Chancellor, A.P.Agricultural University, Rajendranagar, HYDERABAD - 500 030.	Member
5.	Dr. Baidyanath Mishra, Deputy Chairman, State Planning Board, Orissa Secretariat, BHUBANESWAR (Orissa).	Member

6.	Dr. H.R. Kalia, Ex-Vice Chancellor, Neugal Niwas PO Bundla, PALAMPUR - 176 061. (Himachal Pradesh).	Member
7.	Dr. Maharaj Singh, Dy.Director General (Edn.) Indian Council of Agricultural Research, Krishi Anusandhan Bhavan, PUSA, NEW DELHI-110 012.	Member
8.	Shri S.W. Oak, Director (Finance), Indian Council of Agricultural Research, Krishi Bhavan, NEW DELHI - 110 001.	Member
9.	Dr. T.P. Ojha, Director, Central Institute of Agricultural Engineering, Sri Guru Tegh Bahadur Complex, T.T.Nagar, BHOPAL - 462 003.	Member
10.	Dr. K. Mohan Naidu, Director, Sugarcane Breeding Institute, Lawley Road, COIMBATORE - 641 007. (Tamil Nadu)	Member
11.	Dr. R. P. Singh, Director, Central Research Institute for Dryland Agriculture (CRIDA), Santoshnagar, HYDERABAD - 500 659.	Member
12.	Dr. B. Panda, Director, Central Avian Research Institute, IZATNAGAR 243 122 (U.P.)	Member
13.	Sh. Md.Habeebuddin, Asst.Admn.Officer, NAARM, Rajendranagar, HYDERABAD - 500 030.	Member-Secretary

MANAGEMENT COMMITTEE MEMBERS

(L to R) (1) Dr.Hanumanth Rao, (2) Dr.K.Mohan Naidu, (3) Dr.H.R.Kalia, (4) Dr.R.P.Singh, (5) Dr.K.V.Raman, (6) Dr.Maharaj Singh, (7) Dr.A.Appa Rao, (8) Sh.Md.Habeebuddin.

Hindi Day Celebration at the Academy

Dr.K.V.Raman, Director welcomes the Chief Guest Shri Adhyanath Pandey, Faculty, SVP National Police Academy, Hyderabad. Also seen Dr.Devesh Kishore Professor and Chairman, Official Language Implementation Committee.

14. Shri K.L. Aneja, Chief Admn. Officer, NAARM, Rajendranagar, HYDERABAD-500 030. — Member-Secretary (from 16.11.1988)

Steering Committee on Advanced Centre on Agricultural Education and Research Management.

This Committee was initially called the Sub-Project Advisory Committee and has been later renamed as Steering Committee. The 2nd and 3rd meeting of the Steering Committee on the Advanced Centre on Agricultural Education and Research Management were held on January 30, 1988 and August 27, 1988 respectively at NAARM,Hyderabad. The Committee reviewed the progress of the project and provided broad guidelines for its implementation based on the suggestions of user agencies. The following are the members:

1. Dr. K.V. Raman, Director, National Academy of Agricultural Research Management, Rajendranagar, HYDERABAD - 500030. — Chairman

2. Dr. Maharaj Singh, Deputy Director General (Edn.), Indian Council of Agricultural Research, Krishi Anusandhan Bhavan, NEW DELHI-110 012. — Member

3. Dr. N.K. Anant Rao, Retd. DDG(Edn.), ICAR, C/o. Dr.R.J.Galagali, Plot No.84, Sector 12, M.M.Extension, BELGAUM-590 016. — Member

4. Dr. A. Appa Rao Vice Chancellor, A.P.Agricultural University, Rajendranagar, HYDERABAD-500 030. — Member

5. Dr. A.M. Michael, Director, Indian Agricultural Research Institute, NEW DELHI - 110 012. — Member

6.	Dr.R. Nagarcenkar, Director National Dairy Research Institute, KARNAL-132 001.	Member
7.	Dr.A. Ram Mohan Rao, Dean, PG Studies, A.P.Agricultural University, Rajendranagar, HYDERABAD - 500 030.	Member
8.	Dr. K.V. Subba Rao. Professor, NAARM, Rajendranagar, HYDERABAD-500 030.	Member
9.	Dr. Devesh Kishore, Professor, NAARM, Rajendranagar, HYDERABAD - 500 030.	Member
10.	Dr.V.Raghavendra Rao, Professor, NAARM, Rajendranagar, HYDERABAD-500 030.	Member
11.	Dr.Guy Baird, Indian Coordinator, WINROCK International, Support Serveces Contract, Ashok Hotel,Room No.1536, Chanakyapuri, NEW DELHI - 110 021.	Member
12.	Sh.Md.Habeebuddin, Asst.Admn.Officer, NAARM, Rajendranagar, HYDERABAD - 500 030.	Member-Secretary
13.	Shri K.L.Aneja Chief Admn. Officer, NAARM, Rajendranagar, HYDERABAD - 500 030.	Member-Secretary (from 16.11.1988)

STEERING COMMITTEE MEMBERS

(L to R) (1) Sh.Md.Habeebuddin, (2) Dr.K.V.Subba Rao, (3) Dr.Devesh Kishore, (4) Dr.A.Appa Rao, (5) Dr.Maharaj Singh, (6) Dr. K.V.Raman (7) Dr.N.K.Anant Rao, (8) Dr.Guy Baird (9) Dr.A.Ram Mohan Rao.

The Official Language Implementation Committee

The Official Language Implementation Committee of the Academy provided periodical suggestions for implementation of the Official Language Policy guidelines of the Govt. of India issued from time to time. The following are the members:

1.	Dr. Devesh Kishore Professor	Chairman
2.	Dr.K.V. Subba Rao Professor	Member
3.	Shri K.L.Aneja Chief.Admn. Officer	Member
4.	Dr. M.M.Anwer Assoc.Professor	Member
5.	Dr.P.Manikandan Assoc. Professor	Member
6.	Shri D. Venkateswarlu Jr. Hindi Translator	Member-Secretary

The Drafting and Evidence Sub-Committee of the Committee of Parliament on Official Language visited the Academy on 5.2.1988 and reviewed the progress of the implementation of the Official Language.

A training programme to provide working knowledge in Hindi for the employees of the Academy was started on 17.8.1988. Hindi Day was also celebrated on 14.9.88 under the Chairmanship of Dr. K.V.Raman, Director.

9. PERSONNEL

K. V. Raman	Director

Director's Personal Cell

P. P. Krishnan	Sr. Stenographer (upto 8.4.88 AN)
K.S. Sharma	Assistant
T. Srinivas	Jr. Clerk
P. Swamy	SSG I

I. Agricultural Research Systems and Policies Unit

Jagdeesh C. Kalla	Joint Director,Unit in Charge (from 15.12.88 FN)
P. Kumar	Professor (from 4.8.88 FN)
VE. Sabarathnam	Associate Professor
T. Balaguru	Associate Professor
A. Bandyopadhyay	Associate Professor (upto 11.3.88 AN)
K.P.C. Rao	Associate Professor (from 18.3.88 FN)
K. Vidyasagar Rao	Associate Professor (from 4.7.88 FN)
M.J. Pradeep Kumar	Computer Operator
C.V.V.S.S.R.L.Phani Raj	Jr. Clerk
K. Venkateswara Kumar	Statistical Assistant (from 21.9.88 FN)
J.S.R. Murthy	Stenographer

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II. Educational Systems and Technology Unit

Devesh Kishore	Professor, Unit in Charge
M.P. Singh	Professor (upto 14.11.88)
K.M. Reddy	Associate Professor (from 27.8.88 FN)
N.R. Nageswara Rao	A.V. Operator
K. Ajay Kumar	Instruments Technician (from 19.10.88 FN)
M. Venkatesh	Jr. Stenographer

III. Transfer of Technology Systems and Policies Unit

R. K. Samanta	Assoc. Professor, Unit in Charge
R. Chittiraichelvan	Associate Professor
A. U. Gadewar	Associate Professor (from 7.7.88 FN)
L. Venkateswarlu	Photographer
D. Rajagopal Rao	Media Operator (from 2.3.88 FN)
Bansidhar Nayak	Projectionist (from 16.9.88 FN)
M. Kumar Goud	PAE Operator (from 16.5.88 FN)
J. Pramod Kumar	Jr. Stenographer (upto 30.8.88 AN)
A. Vijayalakshmy	Jr. Stenographer

IV. Project Management Unit

V. Raghavendra Rao	Professor, Unit in charge
S.N. Saha	Professor
K. Venkateswarlu	Professor
P.Manikandan	Associate Professor
L. Jhansi Lakshmi	Stenographer.

V. Human Resources Development Unit

K.V. Subba Rao	Professor, Unit in charge
M.M. Anwer	Associate Professor
K. Prathap Reddy	Associate Professor
Sarada Samanta	Stenographer

VI. Information and Documentation Unit

R.B. Gaddagimath	Librarian, Unit in Charge
P.V. Nirmala	Cataloguer
T. Shanmughanathan	Library Assistant
Y.V. Siva Prasad	Documentation Assistant
K. Swarajya Lakshmi	Jr. Library Assistant
Y. Dhanalakshmi	Jr. Clerk

VII. Centralised Services

(i) Farm Operations

V. Murali	Farm Manager
Md. Abdul Basith	Lab Technician
K. Rangaswamy	Field Assistant

B. Veeraiah	Farm Assistant
A. Laxman Maharu	Farm Assistant
K. Prabhudas	Jr. Clerk
K. Thirumalaiah	Pump House Operator
M. Padmaiah	Tractor Driver

(ii) Production and Publication

R.V.V.S. Prakash Rao	Asst. Editor
A. Murali Mohan Rao	Artist
K. Anil Kumar	Reprographic Assistant
K. Desai Babu	Dark Room Assistant (upto 13.4.88 AN)
S. Swamy	Assistant Gestetner Operator
N. Narasimha Reddy	Offset Machine Operator (from 2.5.88 FN)
C. Chandramouli	SSG I

(III) Administration and Accounts

K.L. Aneja	Chief Administrative Officer (from 16.11.88 FN)
G.C. Sharma	Sr.Administrative Officer (upto 15.6.88 AN)
M.P.Chandrasekharan	Accounts Officer
Mohd. Habeebuddin	Asst. Administrative Officer
K. Janaki Ramaiah	Superintendent
S.P. Sastry	Superintendent (upto 8.4.88 AN)
G. Narasimha Reddy	Superintendent (from 26.12.88 FN)
D. Venkateswarlu	Jr. Hindi Translator
T. Ashok Kumar	Assistant

Y. Shankar Rao	Assistant
P.P. Brahmaji	Assistant
A.P. Ramachandran	Assistant
A.P. Jaiswal	Hindi Assistant (up to 5.11.88)
C. Bagaiah	Sr. Clerk
P.G. Kohad	Sr. Clerk
W. Sreenivasa Bhat	Sr. Clerk
M. Narasimha Rao	Sr. Clerk
N. Raghunath	Jr. Steno
M. Dinesh	Jr. Clerk
P. Srinivas	Jr. Clerk
K. Srinivasa Rao	Jr. Clerk
T. Pardhasarathy	Jr. Clerk
G. Raj Reddy	Jr. Clerk
B. Padma Saroja	Jr. Clerk
S. Balakamesh	Jr. Clerk
P.Rajarajeshwari Jyothi	Jr. Clerk
K. Surekha	Jr. Clerk
Syed Ghouse	Jr. Clerk
B. Shashi Vardhani	Jr. Clerk (upto 10.6.88 AN)
P. Sesha Sai Kumar	Jr. Clerk (upto 31.5.88)

Hostel Services

S. Papaiah	Museum Assistant
Sham Bahadur	Catering-in-charge
P. Neelakantam	Sr. Clerk
D. Sangamma	Cook (SSG II)
B. Subba Rao	Asst. Cook (SSG I)
B. Santhamma	SSG I
K. Kalavathi	SSG I
S. Shakuntala	SSI I
G. V. Bikshapathi	SSG I
Phool Kumar	SSG I
A. Yadaiah	SSG I
M. Krishnaiah	SSG I
G. Dasaradha	SSG I
C. Anjaiah	SSG I
T. Jangamma	SSG I

Other supporting services

B. Ch. Satyanarayana	Security Officer cum Caretaker
M.V.V.J. Ravindra Babu	Electrical Overseer (from 5.5.88 AN)
L. Ramesh	Electrician cum Plumber
K. Obulpathi	Technician, Electrical
M. Mohan Rao	Technician, Electrical (from 19.3.88 FN)
M.K. Shamshuddin	Technician, Electrical (from 8.4.88 FN)

B.Satyanarayana Murthy	Technician, Electrical (from 8.4.88 FN)
Y. Venkateswara Rao	Technician, Electrical (from 1.7.88 AN)
A. Nageswara Rao	Lineman (from 10.3.88 AN)
N. Prabhakar Rao	Plumber (from 20.5.88 FN)
Veeranarasaiah	Carpenter cum Painter (from 13.5.88 FN)
K. Sattaiah	Driver
S.K. Peera	Driver
Mohd. Naseer Khan	Driver
G. Muthyalu	Driver
U. Venkataratnam	Driver
N. Ashok	Driver
G. Mani Bai	SSG II
B. Narayana	SSG II
B.H. Dharmaraj	SSG II
P. Bal Raj	SSG I
Khalid	SSG I
M. Yadaiah	SSG I
K. Satyanarayana	SSG I
M. Ashok	SSG I
N. Sukunamma	SSG I
C. Bikshapathi	SSG I
M. Narsing Rao	SSG I

S. Jangaiah	SSG I
B. Bharatamma	SSG I
J. Chandraiah	SSG I
G. Pentaiah	SSG I
M. Yadamma	SSG I
M. Sham Rao	SSG I
P. Eswari	SSG I
M. Eswaramma	SSG I
L. Satyanarayana	SSG I

'The true role of scientists is the discovery of truth, they have no concern either with the application or mis-application of the results of scientific discoveries. All the same, scientists owed an obligation to society, to improve it.'

C.V. Raman

Published by Dr.K.V.Raman, Director, NAARM and printed at NAARM Off-set Press.

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NATIONAL ACADEMY OF
AGRICULTURAL RESEARCH MANAGEMENT
Rajendranagar, Hyderabad-500 030 (India)